Twitter Data Analysis

Analysis of Social Media Activity Regarding the 2018 Mati Wildfire

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Introduction

This report presents a comprehensive computational analysis of a Twitter dataset related to the Mati Fire, a catastrophic wildfire that occurred in the Attica region of Greece in July 2018. The event stands as one of the deadliest wildfires in modern European history, resulting in significant human loss, widespread property destruction, and long-term environmental impact. During and after the disaster, social media platforms—particularly Twitter—played a critical role in shaping public discourse, disseminating real-time information, and facilitating emergency communication and community response.

The dataset under analysis consists of tweets collected over a defined time period surrounding the event and includes key attributes such as anonymized author identifiers, timestamps, tweet text, and engagement metrics (likes, retweets, replies), among others. This rich, multi-dimensional dataset enables the exploration of both temporal and behavioral patterns in online activity during a crisis.

The primary objective of this analysis is to extract meaningful insights regarding user behavior, information propagation, and engagement dynamics in the context of a large-scale natural disaster. To achieve this, a series of computational tasks were performed, including filtering irrelevant content, detecting temporal bursts in tweet activity, analyzing author behavior, examining engagement patterns, and identifying potentially influential or anomalous users.

Beyond technical implementation, this report emphasizes the critical evaluation of methodological choices, assumptions, and limitations inherent in the dataset. The findings aim to contribute to a deeper understanding of how information spreads during emergencies and how social media activity reflects and potentially influences public response during crisis situations.

Dataset Description

The dataset under analysis consists of 289,173 tweets related to the Mati wildfire. The data was initially provided without explicit column names, requiring preliminary inspection to identify and assign meaningful labels to each feature.

The dataset includes the following attributes:

- Author ID
- Timestamp
- Geolocation information
- Tweet ID
- Language
- Like count
- Reply count
- Retweet count
- Source
- Text

These features collectively capture both the content of the tweets and various aspects of user interaction and engagement. It should be noted that not all variables are directly relevant to the objectives of this analysis, and some will be excluded from subsequent stages.

Overall, the dataset provides a sufficiently rich representation of user activity and interaction patterns, enabling both temporal and behavioral analyses. However, as with most real-world social media data, it requires preprocessing and cleaning before meaningful insights can be extracted.

Data Cleaning and Preprocessing

Before performing the main analysis, the dataset underwent a systematic cleaning and preprocessing procedure to ensure that the data handled were both relevant to the event being approached and in a suitable format for analysis.

The first step involved examining the dataset for columns that contributed little or no useful information. This included columns with only a single distinct value, columns where every entry was unique, and columns composed entirely of missing values. These "extreme" columns were removed to simplify the analysis and reduce the size of the dataset, with the exception of the `tweet_id` column, which was retained to allow referencing of original tweets if needed. Additionally, non-extreme columns that were deemed nonessential for the analysis -such as language, geolocation, and quote counts- were explicitly excluded. By removing both irrelevant and extreme columns, the dataset was reduced to a focused set of features most informative for the subsequent analyses, while also achieving a modest runtime improvement.

The tweets' text itself was then processed to standardize and clean its content. All text was converted to lowercase, while mentions, hyperlinks, retweet indicators, hashtags, underscores, and accented characters were removed. Redundant spaces and consecutive punctuation marks were cleaned, and single-character or very short words (fewer than four characters) were filtered out. Emojis and other non-alphanumeric symbols were stripped to focus the analysis on meaningful textual content.

Once the text was cleaned, a keyword-based filtering approach was applied to identify tweets relevant to the wildfire. A curated seed list of keywords related to the event, government figures, disaster response, and affected areas was iteratively expanded by iteratively including frequently co-occurring terms. Each tweet was evaluated against the final expanded keyword set and only those that contained one or more of the keywords list were retained.

```
seed_keywords = [
    "φωτια", "ματι", "πυρκαγια", "νεα δημοκρατια", "συριζα", "τσιπρας",
    "μητσοτακης", "κουλης", "disaster", "νεκροι", "νεκρους", "πυρκαγιες",
    "φωτιες", "lives", "εκλογες", "συριζαιοι", "ζαιοι", "συμμορια", "μαφια",
     "ομαδα", "αληθειας", "δουρου", "ρουφιανοι", "πυροσβεστικη", "προαγωγη",
     "προαγωγες", "θυματα", "κυβερνηση", "ραφηνα", "τραγωδια",
     "κινετα", "ραφηνα", "δικη", "τεμπη", "τρενα", "κατηγορουμενοι", "στελεχη",
     "ξεχασει", "μνημης", "εφημεριδα", "απατεωνες", "ανθρωπινες", "απατεωνες",
     "τυμβωρυχια", "αποζημιωση", "ημερα", "πραγματογνωμονας", "δικαιοσυνη",
     "εκκενωση", "καταθεση", "συγκαλυψη", "μαρτυρια", "πρωτοσελιδο",
     "μαραθωνα", "ανακριτης", "πυροπληκτους", "πυροπληκτοι", "καμμενος",
     "τσουβαλας", "τσουβαλα", "πορισμα", "δικογραφια", "πυρκαγιας", "συγγενεις",
     "πυρκαγια", "ανεμος", "αερας", "αερα", "προστασια", "φλογες", "αγνοειται",
     "περιθαλψη", "τροφιμα", "αναγκης", "αδεσποτα", "χαθηκε", "φαρμακα",
     "τοσκας", "τοσκα", "νομοσχεδιο", "κατασβεση", "περιουσιες", "διαχειριση",
     "πυρκαγιων", "εκατομβη", "παραιτηση", "πυροπληκτοι", "μποφορ", "συγγενων",
     "κακουργημα", "επιτελικο", "μνημη", "ξεχνω", "ξεχαστει", "μαρτυριες", "σιδηροδρομος",
     "επιχειρησιακο", "τροφιμων", "σοροι", "families", "γραμμη βοηθειας", "καηκε",
     "αναζητουνται", "λιστα", "αιμοδοσια", "victims", "αυθαιρετα", "αναγκη", "νοσηλεύονται",
     "ψινακης", "υποκινουμενος", "διασωση", "πτωματα",
]
```

To improve the precision of this filtering process, a list of common Greek stopwords was removed from the text prior to applying the keyword matching. This step ensured that low-information frequently occurring words did not trigger matches, making filtering based on “matching words” more accurate.

```
stopwords = {
    "επειδη", "αυτοι", "γιατι", "εχουμε", "εχει", "είναι", "ειμαστε",
    "στους", "στον", "τους", "παντα", "καμια", "ποιος", "καποιος",
    "εχουν", "ειχαν", "εκανε", "οποιος", "ειχε", "κανουν", "εχουν",
    "στην", "εναν", "ακομα", "ακομη", "αλλα", "αλλη", "αλλο", "αλλος",
    "αμεσα", "αλλιως", "αυτον", "αυτος", "καποια", "καποιοι", "εκτος",
    "εκεινος", "εκεινοι", "ηταν", "τετοια", "τετοιο",
}
```

As a result of this preprocessing and filtering pipeline, 247,955 tweets were retained as the cleaned, event-relevant dataset, while 41,219 tweets were excluded as unrelated or irrelevant. The final dataset contains tweets coming from 28,536 users, spread across a timeline from 23/07/2018 to 07/04/2023.

```
Dataset was cleaned successfully and saved as "mati_clean.csv"
Entries kept: 247955
Entries rejected: 41219

Total authors in the dataset: 28536
Dataset entries spread across 2018-07-23 to 2023-04-07
```

Upon inspecting the rejected tweets, we found that some were indeed relevant to the project. However, other filtering mechanisms which were designed to capture nearly all relevant tweets, were also prone to allowing irrelevant tweets to pass. When weighing these two scenarios, we prioritized minimizing the inclusion of irrelevant tweets over retaining every single relevant one.

This decision was justified for several reasons: first, the number of tweets retained is sufficiently large to support robust analyses; second, the proportion of relevant tweets among those excluded is relatively small, making the impact on overall results minimal.
Consequently, this filtering approach ensured that the final dataset was focused, reliable, and suitable for exploring temporal trends, user behavior, and engagement patterns, while maintaining a manageable level of noise.

Burst Periods Detection

Overview

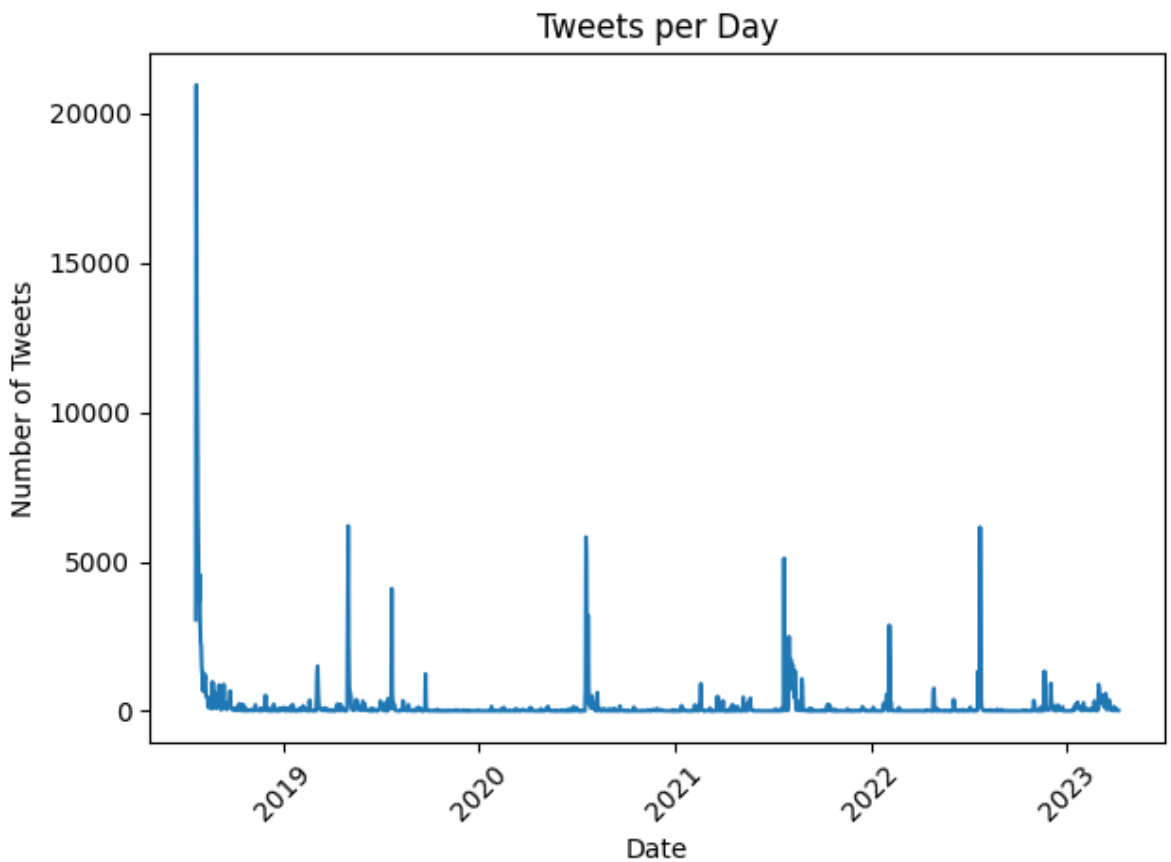
Across the dataset, we observe substantial variation in daily tweet volume, with periods of both very low and extremely high activity. These fluctuations reflect the dynamic nature of public attention, where long stretches of minimal engagement are occasionally interrupted by sharp spikes driven by external events. Identifying these spikes, referred to as *bursts* in this project, constitutes the main objective of this section.

Before proceeding to burst detection, it is important to establish a clear understanding of the raw daily activity patterns and ensure that the dataset is temporally consistent.

Preprocessing and Daily Volume Construction

Tweet activity was aggregated at the daily level, producing a time series of tweet counts per day.

A key preprocessing step was the inclusion of days with zero tweets, which are not naturally present in grouped data. These missing dates were explicitly reintroduced to create a continuous daily timeline. This step preserves the true temporal structure of the data, prevents artificial inflation of averages, and ensures that the rolling statistics introduced in the sequel are computed correctly.



Summary Statistics

The main characteristics of the daily tweet volume are summarized below:

Metric	Value
Peak Day	2018-07-24
Peak Volume	20,958 tweets
Inactive Days (0 tweets)	143 days
Average Daily Volume	144.16 tweets
Median Daily Volume	10 tweets

These statistics highlight an important characteristic of the data: the mean (144.16) is substantially higher than the median (10), indicating a heavy-tailed distribution in which a small number of extreme events dominate overall activity.

This observation reinforces the need for a robust method to distinguish meaningful spikes from typical low-level noise.

Burst Detection Methodology

To identify periods of unusually high activity, we adopt a statistical approach based on rolling averages and variability thresholds.

A key design choice is the size of the rolling window used to compute the moving average and standard deviation. In selecting an appropriate window, two competing requirements must be balanced:

- The window must be large enough so that daily noise does not affect the result.
- It must be small enough so that short-term peaks are not flattened.

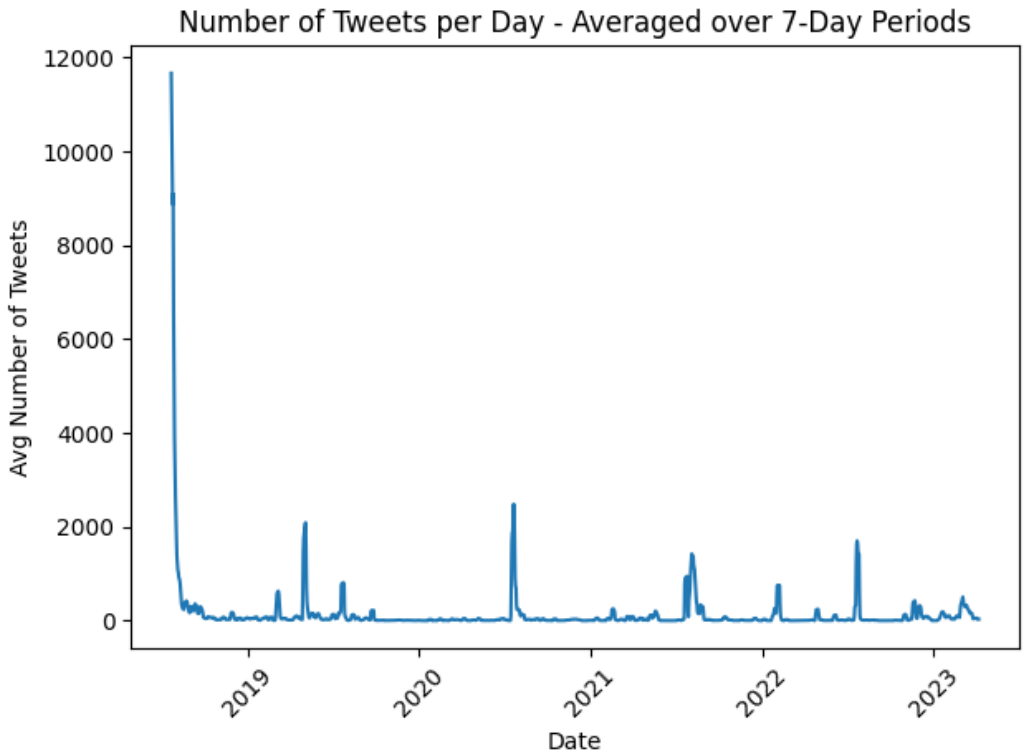
To determine an appropriate balance, we consider the following:

When could noise appear?
Noise can be caused by individual users with many followers tweeting without a major news event having occurred. Such posts may receive interactions (primarily retweets), creating short-term spikes in the data. This activity typically lasts about 2 days [1], while lower levels of engagement may persist for a few additional days. The rolling window should therefore be large enough to prevent this noise from influencing surrounding periods once the effect has faded.

When do short-term peaks occur?
Peaks mainly occur when news stories emerge or during annual anniversaries of the event under study. These periods of increased attention typically span approximately 5–7 days, based on both prior research and empirical observation. The rolling window should therefore be small enough to capture such events without smoothing them out.

Considering both factors, we selected a 7-day centered rolling window. This choice provides an effective balance, as it smooths out short-lived noise while preserving meaningful short-term peaks.

[1] Source: `https://pmc.ncbi.nlm.nih.gov/articles/PMC9362131/`



Having defined the rolling window, we classify a day as a *burst day* if the following conditions are satisfied:

- Its tweet volume exceeds the rolling average plus 1.5 times the rolling standard deviation.
- Its tweet volume is greater than the global median (10 tweets).

The second condition ensures that only statistically significant deviations from typical behavior are captured.

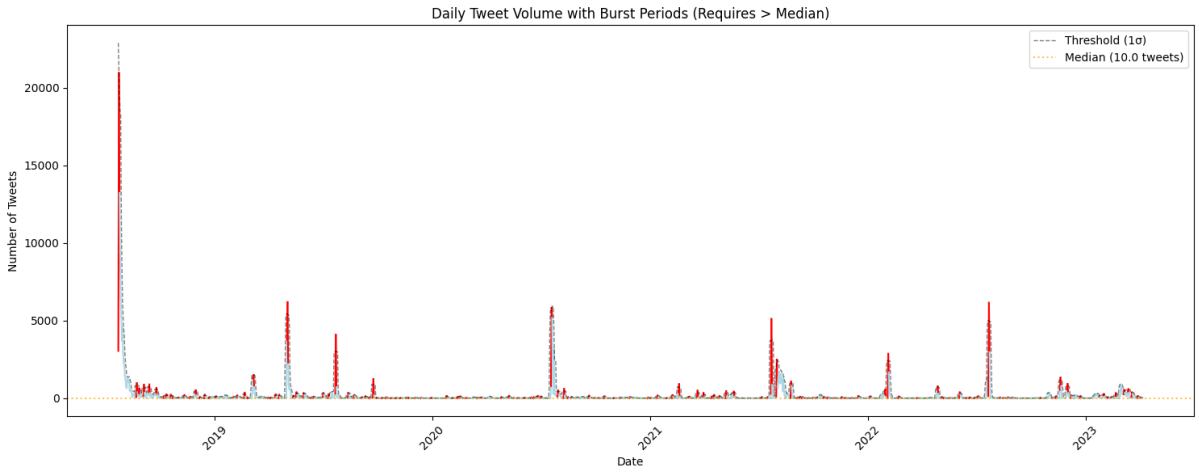
Identification of Burst Periods

Burst days are grouped into *burst periods*, defined as consecutive days for which the burst condition holds.

To identify consecutive sequences, we employed a grouping method based on changes in the burst indicator [2] . Each sequence is assigned a unique Burst Period ID, allowing us to compute:

- The total number of burst periods
- The duration of each burst period
- The average burst duration

[2]Inspired by: `https://www.youtube.com/watch?v=SHvMlFkQAGg`



Results

Metric	Value
Total Burst Days	150
Total Burst Periods	150
Average Burst Duration	1.0 day

The results indicate that all detected bursts correspond to isolated single-day events rather than extended multi-day periods.

Evaluation and Interpretation

The dataset spans approximately 4.5 years, providing a sufficiently long time horizon to assess the robustness and consistency of the proposed method.

Across this period, the burst detection process remains stable, with the adaptive threshold effectively accounting for long-term variations in tweet activity. The absence of systematic over-detection suggests that the method is not overly sensitive to background fluctuations or noise.

A notable outcome is that all detected burst periods have a duration of one day. This result is particularly informative and can be interpreted in multiple ways.

First, it may indicate that attention spikes within this dataset are inherently short-lived. This would suggest that user engagement is primarily reactive and concentrated around specific moments (e.g., breaking news or sudden events), rather than sustained over multiple days.

Alternatively, this pattern may reflect the conservative nature of the detection threshold. By requiring tweet volume to exceed both a dynamic statistical threshold and the global median, the method prioritizes precision over recall, capturing only the most pronounced deviations while potentially excluding weaker but longer-lasting trends.

This interpretation is further supported by the heavy-tailed distribution observed in the raw data, where a small number of extreme events dominate overall activity. In such settings, moderate increases in attention may not be sufficient to surpass the threshold for multiple consecutive days.

Finally, the absence of multi-day burst periods suggests that, while the chosen 7-day rolling window is theoretically capable of preserving sustained events, such patterns are either rare in the dataset or not strong enough to be classified as bursts under the current criteria.

Overall, the method demonstrates a strong ability to isolate clear and statistically significant spikes in activity, making it well-suited for identifying high-impact events, even if it may underrepresent more gradual or sustained increases in attention.

Author Activity

Overview

Among the approximately 28,000 users in the dataset, we observe considerable variability in posting behavior. Some users post very frequently over short periods, others maintain a long-term presence with moderate activity, and many post only sporadically or just once. To systematically capture these differences, we quantify user activity along several complementary dimensions.

Activity Metrics

User activity is characterized in terms of four key aspects:

- **Sustained Presence:** Users who tweet regularly over a long period.
- **Intensity When Present:** Users who post many tweets during periods of engagement.
- **Longevity:** Users who have been part of the platform for an extended time.
- **Frequency:** Users who tweet often, i.e., have small intervals between consecutive posts.

These dimensions are operationalized using the following metrics:

- **Tweet Volume:** Total number of tweets by the user.
- **Activity Duration:** Time span between a user's first and last tweet.
- **Average Interval:** Mean time between consecutive tweets.

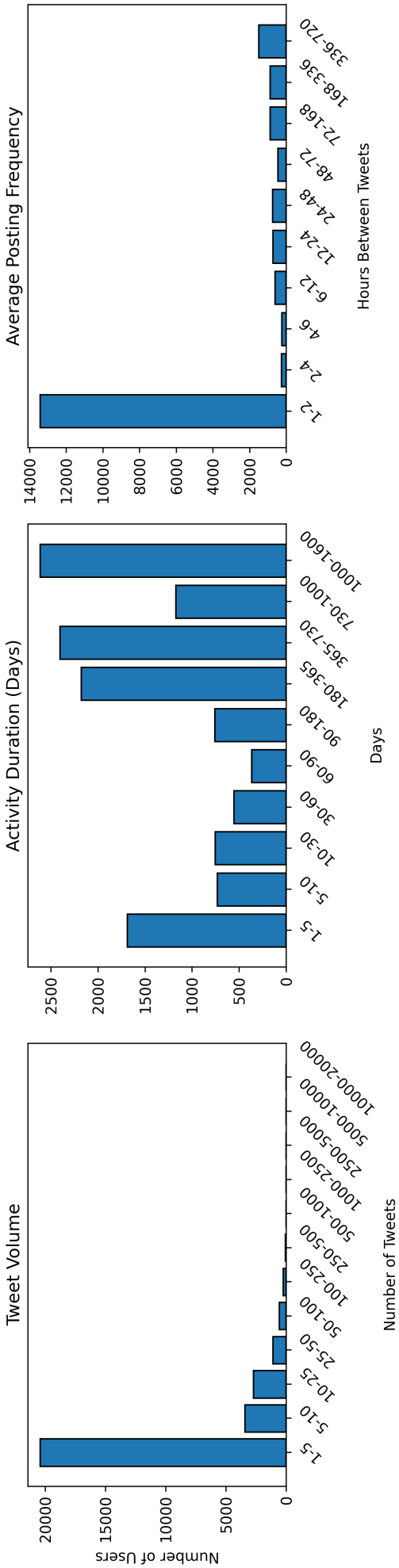
Handling Special Cases

Certain users require special treatment to avoid skewing the rankings:

- **Single-Tweet Users:** These users have no intervals between tweets. Their NaT intervals are replaced with the minimum positive interval observed across all users for fair representation.
- **Zero-Average Interval Users:** We identified three users with zero average intervals, each with exactly two tweets. These extreme cases likely correspond to identical timestamps and non-standard or automated activity. Given their minimal number and negligible contribution relative to the full dataset, they are excluded from the ranking.
- **Very Small Intervals:** Intervals below 1 hour are rounded up to 1 hour to prevent extremely high frequency values from dominating the ranking while maintaining meaningful separation among active users.

Distributions of Activity Metrics

The distribution of user activity metrics reveals the heterogeneity of engagement across the dataset. Most users post relatively few tweets, with a long tail of highly prolific accounts. Activity duration also varies widely: while many users are active over only a few days, some maintain presence for multiple years. Posting frequency shows a similar pattern, with the majority of users tweeting sporadically, and a smaller subset posting at short intervals.
These distributions justify the need for a composite ranking metric that considers tweet volume, activity duration, and posting frequency together, as no single metric alone adequately captures the diversity of user activity patterns.


Tweet Volume
Number of Users
0
5000
10000
15000
20000
1-5
5-10
10-25
25-50
50-100
100-250
250-500
500-1000
1000-2500
2500-5000
5000-10000
10000-20000
Number of Tweets
Activity Duration (Days)
0
500
1000
1500
2000
2500
1-5
5-10
10-30
30-60
60-90
90-180
180-365
365-730
730-1000
1000-1600
Days
Average Posting Frequency
0
2000
4000
6000
8000
10000
12000
14000
1-2
2-4
4-6
6-12
12-24
24-48
48-72
72-168
168-336
336-720
Hours Between Tweets

Ranking System

To combine the activity metrics into a single score, we define a weighted formula:

$$Score = 0.50 \times TweetVolume + 0.25 \times ActivityDuration(days) + 0.25 \times \frac{1}{Frequency(hours)}$$

Choice of Hours for Frequency: Hours are used to keep values interpretable and avoid exaggerating extremely frequent posting behavior.
Weighting Rationale: Tweet volume is prioritized (0.50) because intensity is the primary indicator of activity. Activity duration and posting frequency are weighted equally (0.25 each), reflecting both persistence and consistent engagement.

Next, we present the top 10 most active users according to the composite ranking score, illustrating how intensity, frequency, and longevity combine to identify the most engaged accounts.

Top 10 Most Active Users

author_id	Tweet Volume	Activity Duration	Frequency in Hours	Score
2349489651	1232	1711	33	1044
3173348733	1108	1710	37	982
3300398932	897	1715	46	877
1298248224	858	1702	48	855
305738208	852	1711	48	854
486783894	776	1714	53	817
867807648808407040	745	1710	55	800
4927891804	719	1714	57	788
304830698	671	1710	61	763
250058740	622	1708	66	738

Robustness of Ranking

To validate that the ranking is not an artifact of the specific weight choice, we tested an alternative weighting vector:

- Weighting Vector 1: 0.50, 0.25, 0.25
- Weighting Vector 2: 0.75, 0.15, 0.10

As presented below, the top 10 most active users remain largely consistent under these different weightings. Given that tweet volume is prioritized, the ranking is robust and reflects meaningful activity differences. Minor shifts (the change in the 10th spot in this case) indicate the metric has appropriate sensitivity where it matters. It is important to note that we have not tested extreme weightings where tweet volume is heavily de-emphasized (e.g., weight of 0.1), so this conclusion applies under reasonable prioritization of tweet volume.

Top 10 users with different choices of ranking weight vector:

author_id	author_id (alt)
2349489651	2349489651
3173348733	3173348733
3300398932	3300398932
1298248224	1298248224
305738208	305738208
486783894	486783894
867807648808407040	867807648808407040
4927891804	4927891804
304830698	304830698
250058740	1159260668

Summary

We measure multiple dimensions of activity: intensity, frequency, and longevity.
Outliers (single-tweet or zero-interval users) are carefully handled to avoid distortions.
Binning and plotting allow visual inspection of tweet volume, activity duration, and frequency distributions.
Weighted scoring produces a robust ranking that aligns with intuitive notions of "most active" users while remaining sensitive to meaningful differences.

Time of Day Analysis

Overview

Understanding the temporal patterns of tweet activity provides insights into when users are most and least active throughout the day. This section examines the distribution of tweet activity across hours to identify peak periods, low activity times, and the overall activity range.

Methodology

The hour of each tweet is extracted from its timestamp, and the total number of tweets posted during each hour is aggregated across the entire dataset. To account for the varying length of the dataset timeline, these totals are normalized by the number of days, resulting in the *average number of tweets per hour*.

Formally, for each hour $h \in [0, 23]$:

$$AverageTweets_h = \frac{TotalTweetsatHourh}{TotalDaysinDataset}$$

The following metrics are also identified:

- **Peak Hour:** The hour with the maximum average number of tweets.
- **Lowest Hour:** The hour with the minimum average number of tweets.
- **Activity Range:** The ratio between peak and lowest activity, indicating the relative fluctuation across the day.

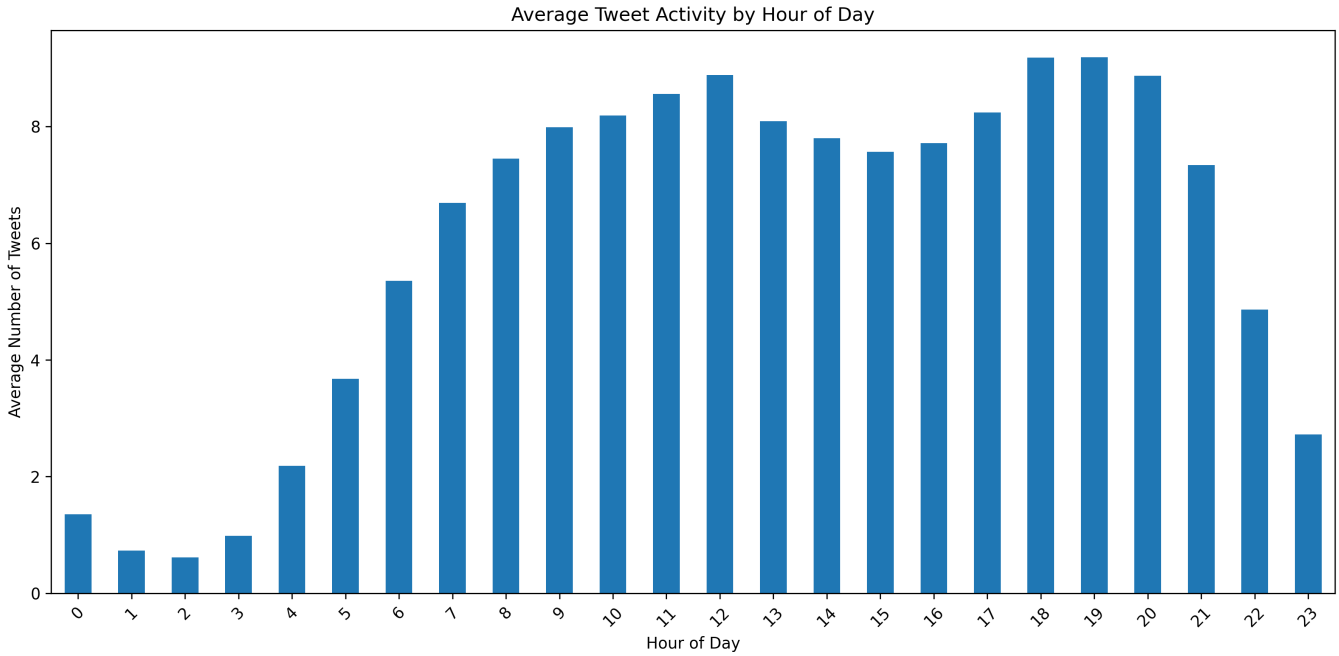
Results

The analysis reveals a clear diurnal pattern in user activity:

- Peak activity occurs at 19:00, with an average of 9.19 tweets per hour.
- Lowest activity occurs at 02:00, with an average of 0.61 tweets per hour.
- The activity range shows that tweet volume at the peak is roughly 15 times higher than at the lowest point.

```
Peak activity: 9.19 tweets at 19:00
Lowest activity: 0.61 tweets at 2:00
Activity range: 14.96 difference
```

This indicates that most user engagement occurs during evening hours, while very few tweets are posted in the early morning. Such temporal patterns can inform time-sensitive analyses, content scheduling, or further investigation into diurnal engagement behaviors. A deeper inspection is facilitated by the bar plot showing the average number of tweets per hour across the dataset.


Average Tweet Activity by Hour of Day
Average Number of Tweets
0
2
4
6
8
0
1
2
3
4
5
6
7
8
9
10
11
12
13
14
15
16
17
18
19
20
21
22
23
Hour of Day

Author-Level Engagement Analysis

Overview

Evaluating engagement at the author level provides insight into how effectively users interact with their audience, beyond simple activity volume. By aggregating interaction metrics and normalizing them per tweet, this section examines how engagement is distributed across users and identifies those who consistently generate strong responses.

Methodology

The dataset is grouped by `author_id`, and for each author the following metrics are computed:

- **Total Likes:** Sum of `like_count` across all tweets
- **Total Replies:** Sum of `reply_count`
- **Total Retweets:** Sum of `retweet_count`
- **Total Tweets:** Number of tweets posted by the author

Using these, additional derived metrics are calculated:

$$AvgRetweetsperTweet = \frac{TotalRetweets}{TotalTweets}$$

$$TotalScore = TotalLikes + TotalReplies + TotalRetweets$$

$$AvgEngagementperTweet = \frac{TotalScore}{TotalTweets}$$

This final metric is used as the primary basis for comparing authors, as it accounts for differences in posting frequency; the normalization ensures that authors are evaluated fairly regardless of how frequently they post.

Results

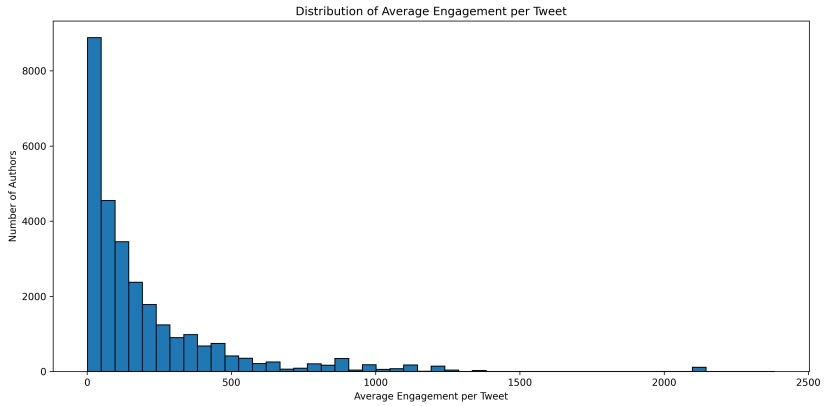
A histogram of the *Average Engagement per Tweet* is used to examine how engagement is distributed across authors.

The distribution is heavily right-skewed, with the vast majority of authors concentrated at low levels of engagement, and the frequency dropping rapidly as engagement increases. This indicates that most users receive limited interaction per tweet, while only a small fraction achieve high engagement.

Beyond this general pattern, an additional detail emerges: while frequencies become extremely low beyond approximately 1500 average engagement, there is a small but noticeable spike just above 2000. Although the number of authors in this range is very small, the presence of a visible bar suggests a localized concentration rather than a smooth decay.

This irregularity may warrant further investigation. It could indicate:

- A small group of highly influential or atypical users



- Inconsistencies of the dataset
- Automated or coordinated activity affecting engagement metrics

For reference, the top 10 authors based on *Average Engagement per Tweet* are presented. The table includes author identifiers, total interaction metrics, total tweets, and average engagement per tweet.

author_id	like_count	reply_count	retweet_count	total_tweets	avg_retweets_per_twee	Total Score	Total Tweets	Avg Engagement Per Twee
320852040.0	248.0	17.0	1059.0	1.0	1059.0	2384.0	1.0	2384.0
765621.0	0.0	0.0	1059.0	1.0	1059.0	2119.0	1.0	2119.0
16922338.0	0.0	0.0	1059.0	1.0	1059.0	2119.0	1.0	2119.0
18339322.0	0.0	0.0	1059.0	1.0	1059.0	2119.0	1.0	2119.0
25910905.0	0.0	0.0	1059.0	1.0	1059.0	2119.0	1.0	2119.0
35083512.0	0.0	0.0	1059.0	1.0	1059.0	2119.0	1.0	2119.0
40455430.0	0.0	0.0	1059.0	1.0	1059.0	2119.0	1.0	2119.0
52781669.0	0.0	0.0	1059.0	1.0	1059.0	2119.0	1.0	2119.0
82611395.0	0.0	0.0	1059.0	1.0	1059.0	2119.0	1.0	2119.0
88900585.0	0.0	0.0	1059.0	1.0	1059.0	2119.0	1.0	2119.0

An important observation from the table is that all top 10 authors have posted only a single tweet, each with identical retweet counts. This results in identical values for *Average Engagement per Tweet*, placing them at the top of the ranking.
While this outcome is mathematically consistent with the metric definition, the uniformity of these values is notable and potentially suspicious. In a natural setting, even highly engaging content typically exhibits some variability in interaction metrics. The presence of multiple distinct authors with exactly the same engagement values suggests that this pattern may not be purely organic.
This pattern strongly suggests the presence of coordinated or automated activity. The exact match in engagement metrics across multiple distinct authors is unlikely to arise from independent user behavior, even in the context of highly viral content. Instead, it points toward scenarios where the same content is systematically propagated across multiple accounts, potentially through bot networks or orchestrated campaigns designed to amplify visibility and interaction metrics.
Such behavior is commonly associated with automated posting mechanisms or coordinated groups

of accounts acting in a synchronized manner. The consistency observed in both tweet count and engagement further reinforces the likelihood that these accounts are not operating independently, but are instead part of a structured pattern within the dataset.

Despite this irregularity, the ranking itself remains consistent with the chosen metric. The table reflects authors with the highest engagement per tweet, rather than the highest total engagement, and therefore naturally favors users with a small number of highly impactful posts.

Time Gap Analysis Between Tweets

Overview

Analyzing the time intervals between consecutive tweets provides insight into user activity patterns beyond simple tweet counts. By examining how frequently users post, it becomes possible to distinguish between consistent behavior, sporadic engagement, and potentially automated activity. This section focuses on identifying irregular posting patterns based on the timing of tweets.

Methodology

The dataset is first sorted by `author_id` and timestamp to ensure chronological consistency. For each author, the time difference between consecutive tweets is computed.

At the dataset level, the median and standard deviation of time gaps are:

- **Median gap:** 12 hours, 11 minutes
- **Standard deviation:** $\sim$131 days

While these statistics provide a general overview, they are not used directly for anomaly detection. The extremely large standard deviation indicates a highly dispersed distribution, likely influenced by long periods of inactivity across many users. As a result, approaches based on global thresholds such as "median $\pm$ 3$\times$std" would not meaningfully capture irregular behavior, as they would be overly permissive and fail to distinguish practically relevant patterns.

Instead, the analysis focuses on interpretable, domain-driven thresholds:

- **Short gaps ($<$ 1 minute):** These are considered indicative of unusually rapid posting, which is difficult to sustain manually and may suggest automated or scripted behavior.
- **Long gaps ($>$ 20 days):** These capture extended inactivity periods. Users exhibiting such patterns align with the notion of *lurkers*, defined as individuals who primarily consume content and only post occasionally.

For each author, the following are computed:

- Average time gap between tweets
- Standard deviation of time gaps
- Number and proportion of short gaps
- Number and proportion of long gaps

Authors are then classified into categories based on these proportions, with additional constraints to ensure sufficient data for reliable conclusions. The specific thresholds and proportion criteria used for classification are documented in the analysis script for reproducibility.

Results

Authors are grouped into four categories:

- **Normal:** No significant irregularities in posting behavior
- **Bot-like:** High proportion of very short gaps (< 1 minute)
- **Lurker-like:** High proportion of long gaps (> 20 days)
- **Mixed irregular:** Combination of both short and long gaps
- **Insufficient data:** Authors with very few tweets (≤ 3), where gap-based analysis is unreliable

Among users with sufficient activity (>3 tweets), 2,526 authors (26.2%) exhibit irregular posting patterns:

- **Lurker-like (1,733 users, 68.6%):** These users are characterized by long inactive periods between tweets, suggesting infrequent and sporadic engagement. A natural question arising from this pattern is whether such users post in bursts when they do become active, an aspect that could be explored further by examining clustering of short gaps following long inactivity.
- **Mixed irregular (551 users, 21.8%):** These users display both very short and very long gaps, indicating highly inconsistent behavior. This may reflect alternating periods of inactivity and intense activity, potentially pointing to seasonal usage patterns or intermittent automation.
- **Bot-like (242 users, 9.6%):** These authors exhibit a high frequency of extremely short gaps, consistent with rapid, automated posting. However, this classification captures only one form of bot-like behavior—namely, high-frequency posting.

Overall, the dominant irregular pattern is lurker-like behavior, while purely bot-like patterns are less common within this framework.

```
Median gap:  0 days 12:11:19
std gap:  131 days 02:12:02.937807
author_type
insufficient_data    18862
normal                7148
lurker_like           1733
mixed_irregular        551
bot_like               242
Name: count, dtype: int64
```

Limitations

It is important to note that this analysis captures only timing-based irregularities. Other forms of suspicious behavior may not be detected by this approach.

In particular, previous extensive engagement analysis identified approximately 100 users with highly unusual patterns: each posted exactly one tweet, receiving zero likes, zero replies, and exactly 1,059 retweets. These accounts are classified here as *insufficient data*, as gap-based analysis requires multiple tweets.

Therefore, the relatively small proportion of bot-like users (9.6%) should not be interpreted as a comprehensive measure of automated activity. Instead, it reflects only one specific behavioral signature. Other forms of coordinated or anomalous behavior may manifest through different signals and require complementary analytical methods.

Summary

Time gap analysis highlights the diversity of user activity patterns. Among authors with sufficient tweets, the most common irregular behavior is infrequent, lurker-like posting, while purely bot-like rapid posting is less prevalent. Mixed irregular users demonstrate alternating periods of very rapid and very slow activity, suggesting inconsistent behavior that may include bursts or seasonal effects.

This approach effectively identifies one type of timing-based irregularity, but does not capture all forms of suspicious or automated activity. As such, the results should be interpreted as indicative of specific behavioral patterns rather than a complete assessment of platform-wide anomalies.

Weekly Activity Patterns

Overview

Analyzing tweet activity across days of the week and weeks of the year provides insight into **temporal engagement rhythms**. This helps distinguish routine posting behavior from activity driven by extraordinary events, and can reveal weekly patterns, anomalies, or bursts of activity.

Methodology

For each tweet, the **day of the week** and the **week of the year** were extracted from the timestamp column. Tweet volumes were then aggregated by week of the year to compute the total number of tweets for each week.

The analysis also separates two periods, based on the timeline of the dataset and prior burst detection analysis:

- **Crisis period:** from the start of the dataset until 31 August 2018, corresponding to the immediate aftermath of the fire (identified via burst detection).
- **Normal period:** the remainder of the timeline, representing typical posting behavior once the immediate crisis had passed.

Tweet volumes were then aggregated by **day of the week** for each period to examine weekly trends.

Observations

Crisis Period

During the crisis period, tweet activity was concentrated on **weekdays**, with much lower activity on weekends:

Day of Week	Tweet Volume
Tuesday	26,117
Wednesday	17,073
Thursday	14,088
Friday	9,833
Monday	9,727
Saturday	7,409
Sunday	5,445

This pattern suggests a **news-cycle effect**, where users engage intensely with breaking news during weekdays, possibly during work or school hours, while weekend activity is comparatively low.

Normal Period

In the normal period, the pattern shifts, with higher engagement on **Fridays, Sundays, and Thursdays**:

Day of Week	Tweet Volume
Friday	27,785
Sunday	25,147
Thursday	24,358
Saturday	23,159
Tuesday	21,513
Wednesday	18,616
Monday	17,685

This suggests a **leisure-time pattern**, where users are more likely to post during free time, evenings, or weekends. Fridays may reflect relaxed engagement as users transition into the weekend.

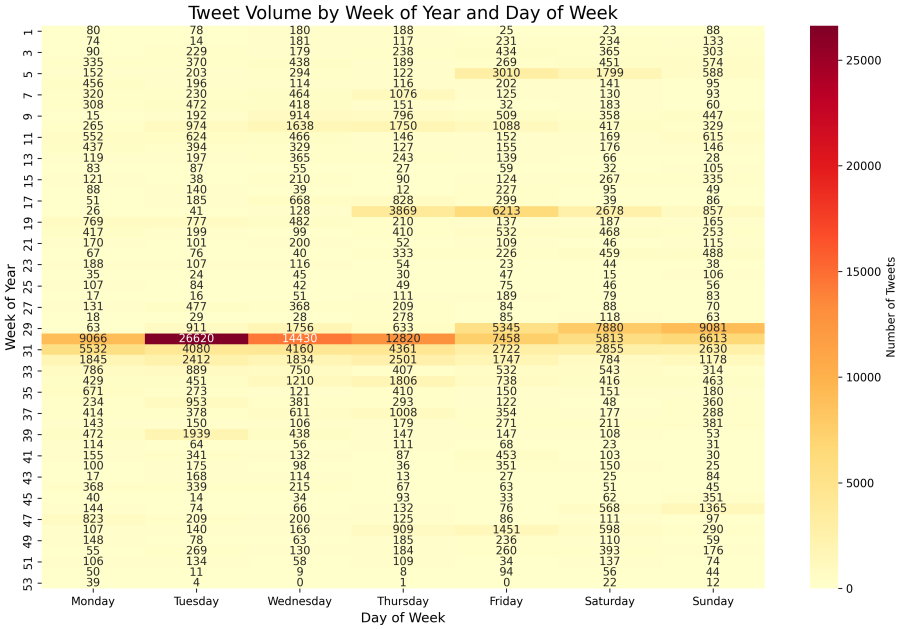
While individual spikes—such as annual anniversary events detected in prior burst analysis—may influence activity temporarily, the **4.5-year span of the dataset** ensures that no single event disproportionately skews the observed weekly pattern. Multiple bursts are distributed throughout the timeline, mitigating the impact of any one anomaly.

```
Crisis period pattern:
day_of_week
Tuesday      26117
Wednesday    17073
Thursday     14088
Friday        9833
Monday        9727
Saturday      7409
Sunday        5445
dtype: int64

Normal period pattern:
day_of_week
Friday       27785
Sunday       25147
Thursday     24358
Saturday     23159
Tuesday      21513
Wednesday    18616
Monday       17685
dtype: int64
```

Heatmap Analysis

The following weekly activity heatmap visualizes tweet volume across weeks of the year (rows) and days of the week (columns), providing a detailed view of temporal engagement patterns. Each cell represents the number of tweets posted on a particular day within a specific week, allowing us to identify both regular rhythms and unusual spikes in activity.



From the heatmap, certain days exhibit extreme activity levels, particularly during the first days following the event. These peaks likely correspond to the immediate response and intense discussion triggered by the incident.

Additionally, activity tends to rise around the anniversaries of the event. Because dates shift across weekdays from year to year (e.g., a specific date may fall on a Friday one year and a Saturday the next), these anniversary-related increases appear on different days of the week each year.

Another notable pattern is the consistent drop in activity during the Christmas period, suggesting that user engagement decreases significantly during major holidays. Outside of these periods—initial event days, anniversaries, and Christmas—there are no other regular intervals showing sustained large changes in activity.

Some days display unexpectedly high activity that does not correspond to the original event or its anniversaries. These spikes may be linked to other major news or developments that temporarily drive user engagement. It is important to note that the dataset spans 4.5 years, which introduces a small artifact: the same weekday and week combination can appear more than once, resulting in periods where consecutive days show roughly double the average activity. While this effect is visible in certain parts of the heatmap, it reflects the temporal overlap of dates across multiple years rather than meaningful changes in user behavior.

Overall, the heatmap provides a comprehensive overview of weekly trends, highlighting both event-driven spikes and regular temporal patterns in user activity.

Author Retweet Dependency

Overview

This analysis examines the extent to which users rely on retweets versus original content. By identifying retweets, we calculate the percentage of tweets that are retweets for each author, determine the retweet-to-original content ratio, and classify authors according to their posting behavior. This helps identify users who primarily amplify existing content versus those who produce original material.

Methodology

Retweets were identified using the `Retweet` indicator in the dataset. For each author, the following metrics were computed:

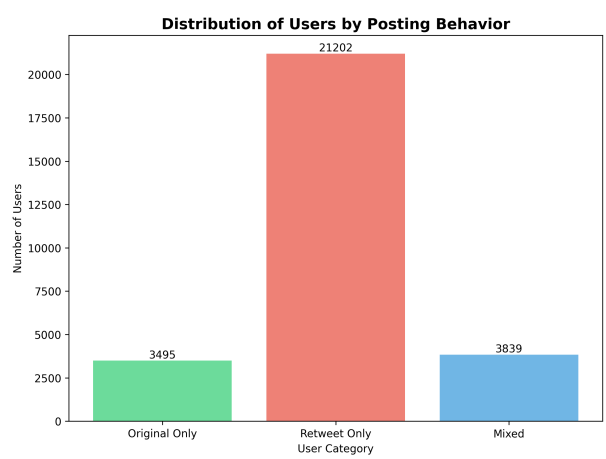
- Total tweets and total retweets.
- Percentage of retweets (`retweet_perc`) and original tweets (`original_perc`).
- Classification into three categories:
  - **Original-only:** authors posting only original tweets.
  - **Retweet-only:** authors posting only retweets.
  - **Mixed:** authors with a combination of original tweets and retweets.
- For mixed users with at least one original tweet, the **retweet-to-original ratio** (`rt_og_ratio`) was calculated.

Results

Total authors: 28,536

User categorization:

- Original-only: 3,495
- Retweet-only: 21,202
- Mixed: 3,839



Top Users

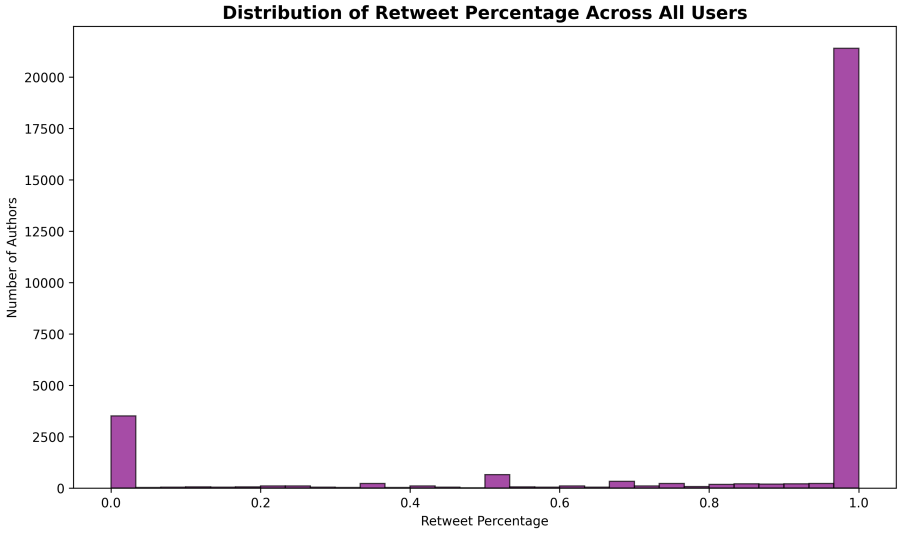
Among mixed users, the highest retweet-to-original ratios are extreme (e.g., 418 retweets for 1 original tweet), showing some users almost entirely rely on retweets.
The top original-only users posted substantial original content, with totals exceeding 600 tweets in some cases.
Top retweet-only users also posted hundreds of retweets, reflecting accounts focused purely on amplification.

Statistics for Mixed Users

- Average retweet-to-original ratio: 2.61
- Median retweet-to-original ratio: 0.167
- Average total tweets per user: 26.89
- Average original tweets per user: 7.44
- Average retweets per user: 19.44

Observations

- Most users are retweet-heavy, with retweet-only accounts dominating the dataset.
- Original-only users are a minority but can post substantial original content.
- Mixed users often have low retweet-to-original ratios, but a small number of users rely almost entirely on retweets, creating highly skewed behavior.



```
Top 10 original-only users (by total tweets):

                    total_tweets  original_count
author_id
196498971                    621             621
17389650                     359             359
2804434455                   355             355
381976833                    331             331
1088823123972186112          188             188
1527074713                   149             149
162385961                    137             137
2893500977                   132             132
25694578                     127             127
3854261175                   123             123
--------------------------------------------------
```

```
Top 10 retweet-only users (by total tweets):

                    total_tweets  retweet_count
author_id
2349489651                  1232           1232
3300398932                   897            897
718905277425324032           533            533
2188942746                   530            530
4263832996                   528            528
2792582272                   506            506
29657174                     445            445
821415519282663427           408            408
816843360                    398            398
3887606853                   396            396
--------------------------------------------------
```

```
Top 10 mixed users by original-to-retweet ratio:
                    total_tweets  original_count  retweet_count  og_rt_ratio
author_id
250058740                    622             621              1        621.0
579792966                    240             239              1        239.0
1271108734476005378          233             232              1        232.0
791389201                    151             150              1        150.0
1583470988                   291             289              2        144.5
563429665                    217             215              2        107.5
370042368                    102             101              1        101.0
373863473                     76              75              1         75.0
2801820046                    64              63              1         63.0
720017205354737664            64              63              1         63.0
--------------------------------------------------
```

```
Top 10 mixed users by retweet/original ratio:

                    total_tweets  original_count  retweet_count  rt_og_ratio
author_id
709816832627253248           419               1            418        418.0
551757163                    419               1            418        418.0
392623428                    357               1            356        356.0
873903623008382977           311               1            310        310.0
601297552                    259               1            258        258.0
2299846784                   248               1            247        247.0
1137408156                   482               2            480        240.0
3046090387                   475               2            473        236.5
2943609773                   218               1            217        217.0
599464925                    205               1            204        204.0
--------------------------------------------------
```

Burst-Origin Analysis

The goal of this analysis is to identify the users who significantly contribute to the early phases of detected burst periods. This complements our burst detection analysis, which operates at the daily resolution and cannot meaningfully capture patterns and user dynamics.

Exploring Early Burst Patterns

To begin, we investigated the temporal patterns of tweets within each burst day. Our objective was to determine **when bursts actually begin**, rather than relying solely on the daily burst detection. While it would have been possible to define bursts at the hourly level by averaging tweet volumes across the dataset and applying a rolling mean over a 7-day (168-hour) window, we opted against this approach for conceptual reasons. Specifically, a multi-day burst period, or even a single-day one, would inevitably fragment into several separate "hourly bursts" due to the natural lull in activity during nighttime hours, when users are inactive, as well as from times during the day when the absolute amount of tweets may decrease, but the trend remains upward. This fragmentation would misrepresent the continuity of burst periods, making it difficult to identify the true initiators of a surge. Instead, by examining the **hour of each tweet** within already identified daily bursts, we could analytically pinpoint the onset of activity while preserving the conceptual integrity of each burst period and accurately identifying the users driving its early stages.

Representative Hourly Activity

As an example, the tweet counts for a representative burst day were as follows:

Hour	Tweets
0	7
1	1
2	4
3	12
4	25
5	35
6	42
7	44
8	69
9	41
10	40
11	52
12	91
13	59
14	49
15	32
16	37
17	73
18	50
19	43
20	37
21	19
22	13
23	12

Inspection of this and other burst days shows that activity begins to rise during the early morning hours, with a clear transition into sustained high activity around 08:00. While we present only this day, the same pattern holds across other burst days in the dataset: several bursts exhibit increasing

activity prior to the main daily peak, which generally aligns with the peak hours identified in the *Time of Day* analysis.
This observation motivates the use of approximately 08:00 as a practical boundary for the onset of burst activity. Activity before this time is typically lower and more irregular, often reflecting isolated or outlier behavior, whereas activity after this point becomes more sustained and representative of broader user engagement. As a result, tweets posted before 08:00 are treated as part of the **early burst phase**, capturing the transition from initial activity to full-scale burst dynamics.

Reasoning for Pre-Burst Day Analysis

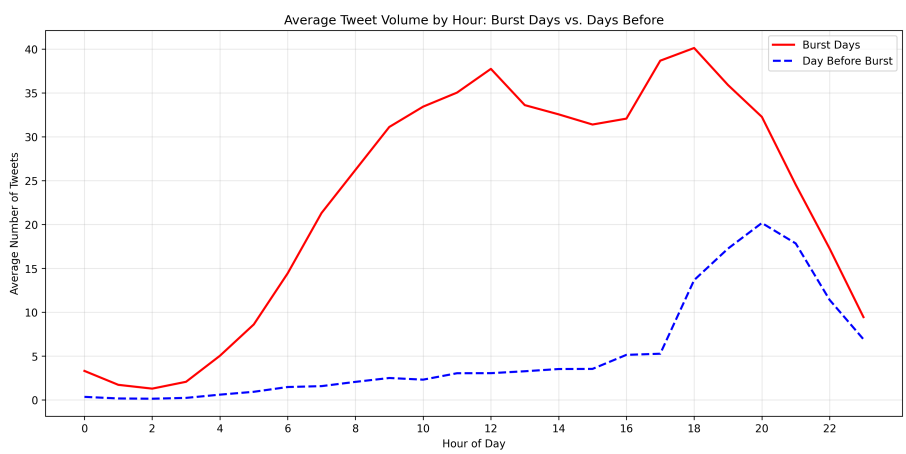
This observation led us to also consider **tweets from the day before each detected burst**. Specifically:

- Early activity often begins around **18:00 on the previous day**, several hours before the first daily peak of the burst day, as shown in the following.
- In some cases, bursts begin on the detected day itself, and in other cases no tweets occur on the preceding day.
- By including the **last evening hours of the previous day** as well as the **early hours of the burst day** (until roughly 8:00), we capture the true onset of the burst rather than only the main peak.

This approach ensures we identify users who either **initiate** the burst or **significantly amplify** it in its earliest phase, rather than mislabeling outliers such as insomniacs or users in different time zones as "burst starters."

To illustrate this early activity, we present a line plot showing **average tweet volume by hour**:

- **Burst Days:** Average number of tweets per hour across all burst days.
- **Pre-Burst Days:** Average number of tweets per hour for the days immediately preceding burst days.



This plot highlights the gradual increase in activity starting on the day before, providing visual confirmation of our reasoning.

Conclusion on Burst Start

From this analysis, we conclude that **burst periods effectively begin on the day preceding the detected burst day**. Including these pre-burst hours is critical for accurately identifying the users who drive the initial activity and early amplification of the burst.

Author Contribution Metrics

Having defined the early burst window, the next step is to identify which users meaningfully contribute to the formation and amplification of burst periods.

```
Found 82,356 tweets from 150 burst days
Found 17,419 tweets from 138 days before bursts
--------------------------------------------------
Found 9627 unique users tweeting in a pre-burst window.
--------------------------------------------------
```

However, a user cannot be characterized as a *burst initiator* solely based on the fact that they posted during an early burst window. Such a criterion would include a large number of users whose activity is incidental rather than influential. Instead, it is necessary to evaluate both the **consistency** and the **impact** of each user's activity.

To this end, we compute the following metrics:

- **Frequency of participation:** the number of tweets posted by the user during early burst periods. This captures whether a user consistently appears at the onset of bursts or only sporadically.
- **Overall engagement:** the total number of likes and replies received by the user's early tweets. This provides a measure of the general visibility and interaction generated by their content.
- **Retweet impact:** the total number of retweets received by the user's early tweets. This metric is treated separately due to its direct relationship with burst formation.
- **Contribution percentage:** the proportion of the user's total tweets that occur within early burst periods, indicating the relative importance of burst-related activity in their overall behavior.

Among these metrics, retweet activity plays a particularly important role. Unlike likes or replies, retweets directly contribute to the overall tweet volume, which is the signal used for burst detection. As a result, retweets are not merely an indicator of engagement, but an active mechanism through which bursts emerge and propagate.

Lag Time Analysis

In addition to measuring activity and impact, it is also important to examine the **temporal positioning** of each user within the burst formation process.

To this end, we compute the **lag time** between a user's first tweet in the early burst window and the defined burst onset (08:00 on the burst day). For each author and burst day, we identify their earliest participation and measure how far in advance it occurred.
Across all author–burst combinations, the lag statistics are as follows:

- Average lag time: 7.54 hours
- Median lag time: 9.72 hours
- Standard deviation: 4.91 hours

- Minimum lag: 0.00 hours
- Maximum lag: 14.00 hours

Percentile analysis further illustrates the distribution:

- 10th percentile: 0.65 hours
- 25th percentile: 1.94 hours
- 50th percentile: 9.72 hours
- 75th percentile: 11.80 hours
- 90th percentile: 13.05 hours
- 95th percentile: 13.50 hours

These results indicate that a substantial portion of users begin contributing several hours before the nominal burst onset, reinforcing the importance of including pre-burst activity when identifying key contributors.

```
Overall statistics across all author-burst combinations:

Average lag time: 7.54 hours
Median lag time: 9.72 hours
Standard deviation: 4.91 hours
Minimum lag: 0.00 hours
Maximum lag: 14.00 hours

Percentiles:
10th percentile: 0.65 hours
25th percentile: 1.94 hours
50th percentile: 9.72 hours
75th percentile: 11.80 hours
90th percentile: 13.05 hours
95th percentile: 13.50 hours
--------------------------------------------------
```

Ranking Burst Contributors

To identify the users who most strongly influence burst formation, we construct a ranking system that combines the previously defined metrics into a single score.

For each user, we consider:

- Early tweet count
- Early engagement (likes and replies)
- Early retweets received
- Contribution percentage
- Number of burst periods participated in

These metrics are aggregated using a weighted scoring function:

- Early tweet count: 35%
- Retweet impact: 35%
- Engagement: 15%
- Contribution percentage: 15%

At first glance, one might argue that the scoring system places disproportionate emphasis on the retweet metric, not because of the raw weight that is assigned to this metric, but because retweet counts are typically of a much larger magnitude than other engagement measures. Consequently, retweets may dominate the final score.

However, this design choice is intentional and reflects the way burst periods are defined in the analysis.

A burst period is primarily identified based on the volume of tweets posted within a given time window. This is the only signal provided directly by the dataset for detecting bursts. Importantly, this tweet volume includes retweets, meaning that retweet activity directly contributes to the identification of a period as a burst.

Moreover, the average lifespan of a tweet is roughly two days, during which it can continue to be retweeted and circulate through the network. Consequently, retweets of posts made during the early phase of a burst period can significantly increase the total tweet volume that eventually triggers the burst detection threshold.

The implication is that retweets are not simply correlated with bursts; they are a primary mechanism through which bursts are formed.

By the time the burst "officially" starts, the users' content who received those retweets, has already been circulating and gaining traction. Those retweets aren't just correlated with the burst - they're accelerating or even causing it.

With this reasoning, keeping retweets as the dominant metric for use score makes perfect sense. The scale difference isn't a bug - it's capturing the actual phenomenon where a single highly-retweeted post can have more burst-initiating impact than dozens of unshared posts.

Top Burst Initiators

To provide a concrete illustration of the ranking methodology, we present the top 12 users according to the burst contribution score.

These users consistently appear in the early stages of burst periods and exhibit strong influence through a combination of high activity and substantial retweet impact. As discussed previously, the prominence of retweet counts in the ranking reflects their direct role in amplifying tweet volume and driving burst formation.

The table below summarizes the key metrics for these top-ranked users.

Author ID	Early Tweets	Engagement	Retweets	Contribution (%)	Score	Bursts
2349489651	126	0	8361	10.23	2971.98	26
1298248224	73	1881	6594	8.51	2616.88	18
3300398932	73	0	6848	8.14	2423.57	18
582026451	96	1469	5182	16.08	2070.06	20
3511406728	75	35	5414	20.78	1929.52	9
305738208	108	9	5305	12.68	1897.80	14
959750025608908800	88	32	5183	18.45	1852.42	16
486783894	83	978	4750	10.70	1839.85	26
885196009596497922	92	82	4982	21.00	1791.35	11
3173348733	79	810	4614	7.13	1765.12	20
718905277425324032	71	0	4940	13.32	1755.85	7
1159260668	64	3364	3321	9.89	1690.83	17

Metric definitions:

- **Early Tweets:** Number of tweets posted by the user during pre-burst windows.
- **Engagement:** Total number of likes and replies received by the user's early tweets.
- **Retweets:** Total number of retweets received by the user's early tweets.
- **Contribution (%):** Average percentage of tweets in the pre-burst window attributable to the user.
- **Score:** Overall ranking score computed from the weighted combination of metrics.
- **Bursts:** Number of pre-burst windows in which the user participated.

Summary Statistics

Across all users and burst periods, the average values of the computed metrics are:

- Average early tweet count: 3.06
- Average early engagement: 13.89
- Average retweets received (early tweets): 283.09
- Average contribution percentage: 32.38%
- Average raw score: 107.09
- Average burst periods participated in: 1.71

These results indicate that while most users participate in only a small number of bursts and contribute limited activity, a smaller subset of users consistently appears in early burst windows and generates disproportionately high levels of retweet activity. These users are therefore the primary drivers of burst formation and early amplification.

```
Averages for all users (across all burst periods):

Average early tweet count: 3.06
Average early engagement: 13.89
Average retweet count "early tweets" received: 283.09
Average percentage contribution (early tweets / total tweets): 32.38%
Average raw score: 107.09
Average burst periods participated in: 1.71
```

Identification of Influential or Manipulative Authors

Overview

All previous analyses naturally lead to a set of important questions: which authors consistently contribute to bursts? Are there users who systematically appear just before bursts begin? Who amplifies information during these periods, and are there patterns suggesting coordinated behavior?

To answer these questions, the analysis in this final section is structured into a series of clearly defined steps, each targeting a different aspect of influence and activity. This structure closely follows the implementation logic of the code and ensures that each dimension is examined independently before combining the results.

Structure of the Analysis:

- **Contribution to Bursts:** Identifying authors who consistently contribute a significant share of tweet volume during burst periods.
- **Catalyst Detection (Pre-burst Activity):** Detecting authors who tend to tweet shortly before bursts begin, potentially acting as triggers.
- **Retweet Amplification:** Measuring whether certain authors experience disproportionate increases in retweet activity during critical periods.
- **Posting Frequency and Temporal Behavior:** Analyzing how frequently and at what times authors post during bursts.
- **Coordination Analysis:** Detecting similarities in posting behavior, co-occurrence patterns, and clustering authors based on temporal activity.

Finally, the results from these steps are combined with previous ranking metrics (engagement, activity, and burst contribution rankings) to evaluate whether the authors contributing to burst initiations also dominate overall activity. Confirming this relationship would strengthen the hypothesis that a subset of users plays a prominent role in sustaining and shaping the conversation on the platform.

Contribution to Burst Activity

To identify consistently influential authors:

- Tweets were filtered to include only those occurring during detected burst periods.
- For each burst day:
  - The number of tweets per author was computed.
  - Each author's percentage contribution to total daily tweet volume was calculated.
- Authors appearing in at least 30 burst periods were considered **consistent contributors**.
- Their average contribution percentage across bursts was used as a measure of sustained influence.

This approach highlights users who repeatedly contribute a non-trivial share of activity during bursts.

Results

For reference, the following table presents the top 10 authors ranked by their average contribution percentage during burst periods.

Author ID	Avg. Contribution (%)
1244628291404824578	1.82
250058740	1.63
2349489651	1.45
179423854	1.39
29657174	1.38
371664114	1.38
779730812786839552	1.36
582026451	1.35
809406596736491520	1.26
711519314684289024	1.24

Observation: The contribution percentages remain relatively low (approximately 1%–2%) when compared to the average contribution pct during a pre-burst window, which is expected given the substantial increase in total tweet volume during bursts. As activity intensifies, the relative share of even highly active users becomes diluted.

Additionally, the next table presents the top contributors ranked by the number of burst periods they participated in.

Author ID	Burst Count	Avg. Contribution (%)
2349489651	73	1.45
486783894	62	1.11
3173348733	54	1.24
1298248224	54	1.15
867807648808407040	54	1.19
4927891804	52	0.97
305738208	52	0.92
711519314684289024	48	1.24
779730812786839552	44	1.36
1244628291404824578	44	1.82

Observation: Several authors appear in more than 50 burst periods, indicating sustained and repeated participation. This suggests the presence of a core group of highly active users who consistently engage during high-activity phases.

Identification of Catalyst Authors

To detect authors who may act as burst triggers:

- Pre-burst windows identified in the previous analysis were examined using lag time metrics.
- For each author:
  - The average time between their first tweet and the start of a burst was computed.
- Authors were filtered to those appearing in at least two pre-burst periods to ensure consistency.

Authors with very small lag times tend to tweet immediately before burst onset, suggesting that they may act as **early contributors or potential catalysts** of increased activity.

Results

Figure below illustrates the top authors sorted by their average lag time relative to burst initiation. These users consistently appear close to the onset of bursts, indicating a possible role in triggering or accelerating activity spikes.

```
Potential catalyst authors (close to burst starts and consistent)

                author_id  first_tweet_lag_hours  burst_count
63    933366882979209221                   0.09            2
126           2856478103                   0.19            2
154  1291243536374005760                   0.22            2
158             60537888                   0.22            2
223            306113154                   0.29            2
224            722922079                   0.29            3
225            231816425                   0.29            2
248            426755433                   0.32            2
328            311877937                   0.41            2
332           3346987277                   0.41            2
--------------------------------------------------
```

Rationale for Excluding Burst Periods

This part of the analysis focuses strictly on pre-burst activity, excluding the burst itself. This decision was made for three key reasons:

- **Isolates early activity:** By focusing only on pre-burst windows, we measure which authors are already influential before the burst begins, helping to identify potential catalysts.
- **Cleaner attribution:** Excluding the main burst prevents conflating early influence with general burst dynamics, ensuring that high retweet counts are due to the author's activity rather than the burst itself.
- **Avoids hype effects:** During bursts, many authors receive retweets simply because overall activity is high. By excluding the burst, we avoid mistaking this general amplification for individual influence.

Retweet Amplification Analysis

To identify authors who amplify information during critical periods:

- Baseline engagement was computed as the average number of retweets per tweet.
- For pre-burst windows:
  - A new metric, **burst retweets per tweet**, was calculated.
- A **retweet boost ratio** was defined as:

$$retweetboost = \frac{burstretweetspertweet}{averageretweetspertweet}$$

To ensure robustness, authors were filtered using the following criteria:

- At least 3 tweets in pre-burst windows
- Participation in at least 2 bursts
- Non-zero baseline engagement

This approach identifies authors whose content becomes disproportionately amplified during periods of increasing activity.

Results

Below the top authors ranked by their retweet boost during burst periods are presented.

Author ID	Avg. Retweets/Tweet	Burst Retweets/Tweet	Retweet Boost	Early Tweets
992148090558836739	6.54	62.33	9.53	9
1258025708401692672	8.06	65.33	8.11	6
1424448930281103362	17.04	111.67	6.55	3
3061245947	27.24	163.33	6.00	3
2539642129	7.63	40.50	5.31	4
563429665	0.87	4.40	5.05	10
2153079170	31.21	151.80	4.86	5
291841627	1.31	6.33	4.82	3
1088823123972186112	0.09	0.38	4.41	8
20305098	22.25	87.81	3.95	21

Observation: Several authors exhibit extremely high amplification ratios (up to ∼9x), indicating that their content spreads significantly more during critical periods compared to normal conditions.

This suggests that these users may play a key role in **information propagation**, even if they are not necessarily the primary initiators of bursts.

Temporal Posting Behavior

The next step is to identify authors who post frequently during bursts and examine their temporal patterns. To achieve this, we computed the **average number of tweets per burst day** for each of the 1918 authors filtered to serve a significant contribution to burst initiations (ie, to belong in at least one of the thre previous groups: burst contributors, catalysts, and "propagators"). This metric allows us to identify users who dominate activity during burst periods and may act as major amplifiers.
Below are presented the top authors based on posting volume on burst periods.

Top authors by average posting frequency:

Author ID	Avg tweets per burst day
1456211606774427658	61.33
938173363939405829	47.00
826906463100628993	46.00
3698751154	36.00
3338719816	21.00
934788201251770375	16.25
942787141989490689	13.00
1413199701902598148	13.00
332771115	13.00
271289371	13.00

Similarity and Coordination Between Authors

The next step is to detect coordinated behavior among authors. In order to begin searching for coordinated efforts of initiating a spike in activity, for each pair of authors, we computed the **temporal similarity of their posting patterns** based on hourly distributions during bursts. Specifically:

- For each author, we built a 24-dimensional vector where each entry represents the fraction of tweets posted in a given hour (0–23) across all bursts. Each vector was normalized so that the sum of entries equals 1.
- Temporal similarity between authors was then computed using the **cosine similarity** between their vectors.
- Authors with a similarity above **0.9** were considered *highly similar* and potentially coordinated.

Highly similar author pairs (similarity $\geq$ 0.9): found **1092** pairs.

Examples of top pairs:

Author 1	Author 2	Similarity
14234604	4063221322	0.92
16874725	47675396	1.00
16874725	752586365968211968	0.91
18392747	206599445	0.90
18392747	951384998279598080	0.90
18899147	875005237	1.00
18939033	30701547	1.00
18939033	46085589	1.00
18939033	80883060	0.90
18939033	112308292	1.00

Clustering Authors by Temporal Patterns

The next step is to group authors based on their hourly activity patterns to detect distinct behavioral clusters. To this end:

- Each author's hourly tweet fractions (0–23) were used as input features for clustering.
- A clustering algorithm (K-Means) grouped authors with similar daily posting rhythms.

- Clusters reveal different temporal strategies, including concentrated activity at specific hours that may indicate coordinated or automated behavior.

Cluster distribution:

Cluster	Number of authors
3	771
4	722
0	215
2	119
1	68

Coordinated Pairs and Connectivity

To better understand potential coordination among authors, we analyzed the temporal alignment of their activity during bursts, with the goal of identifying coordination networks. The methodology can be summarized as follows:

- For each pair of authors with a similarity of at least 90%, we counted the number of times they co-appeared in the same hour-window within the same burst period.
- Only pairs that co-occurred in at least three bursts were retained, to reduce noise and avoid random coincidences.
- For all authors involved in these pairs, we computed the number of coordinated pairs each author belonged to, defining the *degree of connectivity* for each author.

The top authors for each category are presented below, for reference.

Top coordinated author pairs (most frequent co-tweeting first):

```
                                   author_pair  co_occurrences
14752          (2349489651, 867807648808407040)              62
14080           (582026451, 867807648808407040)              55
2686                  (2349489651, 3173348733)              54
8499                   (582026451, 2349489651)              54
11397                 (1298248224, 2349489651)              49
36342          (1298248224, 950647208017350656)              49
36832  (867807648808407040, 950647208017350656)              48
15296          (4927891804, 867807648808407040)              47
35894          (2349489651, 950647208017350656)              47
37792           (582026451, 950647208017350656)              47
```

Author connectivity (number of coordinated pairs each author is in):

	num_coordinated_pairs
2349489651	1118
867807648808407040	962
1298248224	901
582026451	879
3173348733	876
305738208	837
4927891804	818
1159260668	755
486783894	747
304830698	721

Overall, this analysis highlights a subset of highly connected authors who frequently co-occur in bursts, suggesting they may play a central role in coordinated amplification within the network.

Finding Influential Authors

To further identify authors who may exert the most influence in coordinated activity, we combined the results of the previous coordination analysis with overall activity levels. Specifically, we focused on authors who consistently appeared in coordinated pairs across a substantial portion of burst days.

- We retained only pairs with at least **30 co-occurrences**, representing coordination in approximately 20% of total burst days.
- From these pairs, we extracted all unique authors involved in strong coordination.
- Authors with a *connectivity* of at least 200 (participating in at least so many "coordinated pairs") were considered **highly connected**, highlighting central actors in the network.

Applying these criteria, a total of **38 authors** were identified as both strongly coordinated and highly connected, forming the core group of influential participants in orchestrated bursts. These authors are likely to play a central role in amplifying content across multiple bursts and coordinating activity within the network.

Validating Influence of Authors Found

To assess whether the authors identified as highly coordinated also hold influence in the broader dataset, we compared their rankings across several global metrics:

- **Burst contribution rank:** Position of each author based on their role in initiating or amplifying bursts, computed across all bursts.
- **Total engagement rank:** Position of each author according to overall engagement metrics (e.g., likes, retweets, replies) across the entire dataset.
- **Activity rank:** Position based on overall posting activity, capturing how active each author is regardless of coordination patterns.

By combining these rankings, we obtain a multidimensional view of influence, highlighting authors who are not only central in coordination networks but also highly active and impactful overall.

The results are summarized in the table below.

Author ID	Burst Contribution Rank	Total Engagement Rank	Activity Rank
2349489651	1	3	1
1298248224	2	1	4
3300398932	3	5	3
582026451	4	19	12
305738208	6	7	5
959750025608908800	7	24	23
486783894	8	6	6
3173348733	10	2	2
1159260668	12	4	13
4248471437	14	21	15
711519314684289024	15	26	16
867807648808407040	16	8	7
2574632838	17	40	35
3046090387	18	27	24
4927891804	19	9	8
342622445	20	15	14
2188942746	21	11	17
2440608086	22	22	32
29657174	23	59	28
3371205051	26	127	69
3033217867	28	82	31
1009374470908383233	29	80	62
950647208017350656	32	10	25
252807721	34	46	47
304830698	35	28	9
315927762	44	36	49
4263832996	45	16	18
839530553900732416	49	30	20
392623428	60	57	40
999192747973234688	79	203	105
1137408156	89	18	22
1390044422	93	41	41
1609120302	137	187	81
1131567840627240963	165	49	66
779730812786839552	182	20	29
66699630	454	148	34
309668689	939	121	64
196498971	3902	1795	11

Overall, this analysis confirms that the subset of **38 highly coordinated authors** consistently ranks near the top across multiple metrics. This reinforces their influence both in coordinated bursts and in broader engagement and activity patterns, validating the significance of the coordination signals identified in earlier analyses.

Discussion and Conclusions

The analysis of coordinated pairs and author connectivity reveals a small subset of accounts that are disproportionately central. These authors not only exhibit persistent temporal alignment with one another but also rank highly across multiple metrics, including contribution, total engagement, and overall activity. While high activity and engagement alone could indicate genuine influence, the combination with repeated, tightly synchronized posting patterns strongly suggests orchestrated amplification rather than purely organic behavior.

The temporal alignment observed is unlikely to occur by chance given the frequency and persistence of co-occurrences. Moreover, the network of highly connected authors forms a core cluster that repeatedly appears across the dataset, a pattern indicative of deliberate coordination designed to maximize visibility. While some of this activity could be explained by simultaneous responses to trending events, the degree of synchronization, breadth of connectivity, and repeated patterning point toward intentional coordination.

Combined with suspicious activity come across the dataset, such as the finding of multiple users with a single tweet, all of which received exactly 1,059 retweets, as well as the characterization of almost 10% of the dataset authors as "bot-like" based on the frequency of their posts, a clear picture emerges: the dataset is dominated by groups of users operating in unified and coordinated manners, if not automated in certain cases.

Examining the scores across various ranking categories for the users that form the core of these groups confirms that their strategy is effective: the accounts that contribute most to these groups and occupy their core dominate the top ranks across multiple metrics. Among those users, we would not be surprised if we found one with a username inspired by a Soviet General of World War II.

In conclusion, the evidence strongly points toward orchestrated activity as a dominant force shaping visibility and engagement within the dataset. While genuine influence is undoubtedly present among some users, the combination of persistent coordination, extreme amplification events, and bot-like behavior indicates that strategic, collective action drives a substantial portion of the observed activity. Recognizing these patterns is crucial for understanding how attention and reach are manipulated in online networks and for informing further analyses aimed at distinguishing organic influence from coordinated amplification.